

Diagnostic Procedures and Decision Logic

Stamping press 01 (press-stamp-01) | Main drive and lubrication circuit

Machine	Stamping press 01 (machine ID press-stamp-01, type press, area Press shop, site vit-lab), stamping press in a car plant
Sensors	MAIN_MOTOR_CURRENT (A), LUBE_OIL_PRESSURE (bar), BEARING_VIBRATION_RMS (mm/s) and the 64-bin spectrum BEARING_SPECTRUM (mm/s); sensor IDs PRESS-STAMP-01.<CHANNEL_KEY>
Components / subsystem	Press main drive and its pressure lubrication circuit: main drive motor, lubrication unit (gear pump, reservoir, lube filter) and main bearing housing; planned agent tags main_drive_motor, lube_filter, lubrication_unit, main_bearing_housing
Document title	Diagnostic Procedures and Decision Logic
Document ID	PS01-DOC-03
Revision / date	1.0 / 2026-10-08
Purpose	Provides the structured procedure that takes an abnormal telemetry observation of press-stamp-01 to a likely fault, severity, escalation, corrective action and post-maintenance verification, with explicit IF/THEN rules and an evidence-weight matrix.
Scope	Fourteen procedure steps for the three scalar channels and the bearing spectrum. Level definitions are in Document 01, signatures in Document 02, maintenance in Document 04 and worked cases in Document 05. Only the two simulated faults carry project signatures; all other candidates are engineering reference.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_PRESS_STAMP_01.md (catalog limits, simulation equations, topics, runbook stubs, tests).
[D]	Derived value: computed from the repository's own simulation model (seed 1, 2 Hz, 600 s per case) or algebraically from its equations, as recorded in the context file.
[E]	Engineering reference: general industrial practice for press drives, lubrication circuits and vibration diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. There is no physical press or transducer. An ESP32 gateway (or a software simulator) plays the press controller and generates values from documented equations. The repository describes its limits as illustrative constants chosen for the demo, not vendor values, and its runbooks as not an OEM manual. Values tagged [P] are therefore prototype values, not OEM specifications; values tagged [E] and [I] are not manufacturer-certified.

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1. Prerequisites and Feature Definitions

1.1 Inputs

Table 1. Inputs to the procedure

Input	Source	Rate	Tag
MAIN_MOTOR_CURRENT, LUBE_OIL_PRESSURE, BEARING_VIBRATION_RMS	telemetry topic (m object)	500 ms	[P]
BEARING_SPECTRUM, 64 bins	spectrum topic	2000 ms	[P]
status, seq, ts, synced	status, telemetry	each message	[P]
diag (uptimeS, rssi, heap, reconnects)	diag topic	15 s	[P]
ack of cmd/scenario	ack topic	on command	[P]
Machine state (running, stopped, setup)	Not available in the repository; operator or controller	n/a	[E]
Baseline values	Document 01 healthy values or a recorded baseline	per repair	[D] / [I]

The scenario label and the ramp r are ground truth held in diag.sim and served only by a protected endpoint. The agent must not use them. The procedure reasons from channel evidence only [P].

1.2 Feature computation

Table 2. Feature definitions [D] / [I]

V10 = mean(BEARING_VIBRATION_RMS over last 10 s)	V60 = mean over last 60 s
P10 = mean(LUBE_OIL_PRESSURE over last 10 s)	P60 = mean over last 60 s
I60 = mean(MAIN_MOTOR_CURRENT over 60 s)	Ipk = max over each 4.6 s stroke; use 3 strokes (13.8 s)
PF60 = share of samples in 60 s with V >= 2.5	
HB300 = mean(amp[24..63]); PMR300 = max(amp[24..63]) / median(amp[24..63])	
DL = amp[b] / healthy[b] for b in 7, 14, 21, 28, 35	
RMSchk = V - sqrt(sum(amp^2)) / V	
dI/dV = (I60_now - I60_base) / (V60_now - V60_base) (A per mm/s)	
r_clog_P = (4.50 - P60) / 2.03 ; r_clog_V = (V60 - 1.31) / 3.68	
r_bear_V = (V60 - 1.31) / 1.58 ; r_bear_I = (I60 - 94.8) / 7.6	

Window lengths follow the 4.6 s stroke and the 1 Hz pump pulsation: a 10 s window holds about two strokes, a 60 s window about thirteen. The 2 Hz publication rate gives 20 and 120 samples respectively.

2. Procedure Overview

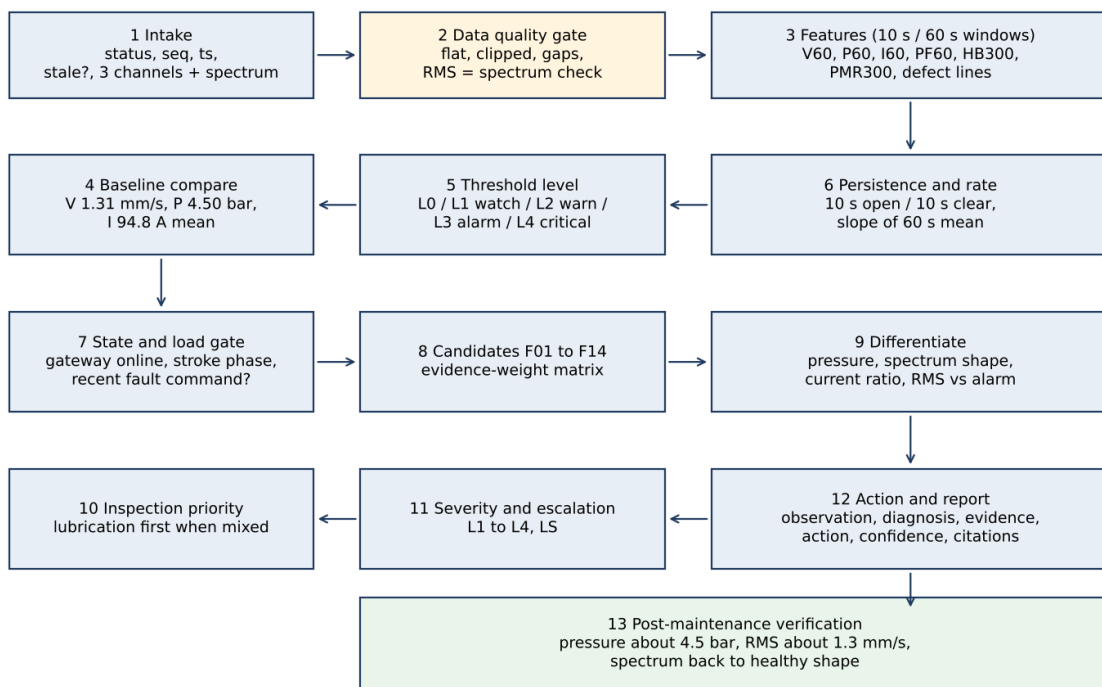


Figure 1. Diagnostic workflow from intake to post-maintenance verification (steps 1 to 13; step 14 is the report). Prototype logic [1].

Table 3. Fourteen-step procedure

Step	Name	Question answered	Output	Section
1	Initial telemetry assessment	Is the data valid and is the machine running?	Valid / LS / suppressed	3
2	Baseline comparison	How far is each feature from its own baseline?	Deviations, ratios	4
3	Threshold evaluation	Which level does each channel reach?	L0 to L4 per channel	4
4	Persistence and duration	Is the level held for 10 s?	Opened or cleared level	5
5	Rate of change	How fast is it moving, and is it drift?	Slope, time to alarm	5
6	Machine operating state	Is the state valid for diagnosis?	State gate	6
7	Load conditions	Is the change confined to the working stroke?	Load attribution	6
8	Candidate fault generation	Which faults fit?	Candidate list	7
9	Fault differentiation	Which candidate is best supported?	Scores, margin	8
10	Inspection prioritisation	What is checked first?	Ordered inspection list	9
11	Severity classification	How serious is it?	S1 to S4	9
12	Escalation	Who must act and by when?	Notification	9
13	Corrective action	What should be done?	Action	10
14	Post-maintenance verification	Did it work?	Pass / fail	10

3. Steps 1 to 3: Assessment, Baseline and Thresholds

3.1 Step 1: initial telemetry assessment

Table 4. Initial assessment rules

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-01	status is not online, or no telemetry for max(5 s, 4 x intervalMs)	LS data state	gateway diag (uptimeS, rssi, mqttReconnects), status topic	F15 data-path fault	No mechanical diagnosis; follow the data-path table
R-02	bootId changes with seq back at 0	LS information	uptimeS, reset reason, recent cmd/reboot	Gateway reboot (a simulated fault is cleared by cmd/reboot)	Restart the window and recompute the baseline
R-03	Standard deviation of a channel is near zero over 60 s while the others vary	LS	Healthy P has 0.03 bar noise and 0.12 bar ripple; I and V breathe with the stroke	F12 (P), F13 (V) or F14 (I)	Inspect the sensor against a reference
R-04	Received RMS differs from the square root of the summed squared spectrum bins by more than 2 percent [I]	LS	Latest spectrum within 2 s; decoding	F13 or F15	Resolve data before any mechanical diagnosis
R-05	An ack with ok true for cmd/scenario was seen in the window	Injected simulation event	ack detail, cmdId	Operator fault injection	Record as injected; do not escalate as a plant fault
R-31	Machine stopped, in setup or die change [E]	Suppress	State input	No limit-based diagnosis	Wait for production

Order matters: data and sensor questions are answered before mechanical ones, because a flat signal or a stale device mimics a fault. The repository's data-path table (Document 04) lists the likely causes of each data symptom.

3.2 Step 2: baseline comparison

Table 5. Baseline comparison table [D] / [I]

Feature	Baseline	Watch ratio or band	Flag as drifting
V60	1.31 mm/s	at or above 1.15 x (1.50 mm/s)	Rising 0.3 mm/s in 10 minutes
P60	4.50 bar	at or below 0.955 x (4.30 bar)	Falling 0.3 bar in 10 minutes
I60	94.8 A	at or above 1.05 x (99.5 A)	Rising 2 A in 10 minutes
IpK	about 150 A	above 160 A	Three strokes in a row
HB300	0.030 mm/s	at or above 0.12 mm/s	Rising
PMR300	not specified in repository	above 2.5 and rising	Above 3 with defect bins

- Use the channel's own baseline, not a fleet average; the repository states a steady climb from 1.3 towards 3 mm/s is a drift whatever the limit says [P].
- Recompute the baseline after any repair, die change or oil change (Document 01 Section 19).

3.3 Step 3: threshold evaluation

Table 6. Threshold classification rules

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-06	P10 at or below 3.6 bar for 10 s	L2 warning	V60, HB300, PMR300, I60 ratio	F01 candidate (also F03, F04, F06, F12)	Run differentiation; replace filter if F01 confirmed
R-07	P10 at or below 3.0 bar for 10 s	L3 alarm	V60 at or above 4.0 mm/s and a broadband floor	F01 severe, or F03, F04	Intervene this shift; filter swap if broadband
R-08	P10 at or below 2.5 bar for 10 s [I]	L4 critical	V10, I60, state	Oil starvation of the bearings	Controlled stop and inspect
R-09	V10 at or above 2.5 mm/s for 10 s	L2 warning	P60, PMR300, HB300	F01 if P60 falls, F02 if P60 normal with lines	Lubrication first; inspect at next stop
R-10	V10 at or above 4.0 mm/s for 10 s	L3 alarm	P60 at or below 3.0 bar?	F01 (bearing wear never reaches 4.0 mm/s in the model [P])	Filter swap now; inspect bearings afterwards
R-11	V10 at or above 7.0 mm/s for 10 s [I]	L4 critical	Mount and cable, impacts, other channels	Severe fault or F13	Controlled stop; check sensor
R-12	Ipk at or above 175 A in 2 of 3 strokes	L2 warning	Idle level 88 A? V60 and P60 normal?	F09 process or F08 motor	Check die, process, drive
R-13	Ipk at or above 195 A in 2 of 3 strokes	L3 alarm	Stroke profile, drive protection	F09 or F08 severe	Reduce load; inspect
R-14	Ipk at or above 220 A, or a sample at or above 240 A [I]	L4 critical	Jam or short	Overload or electrical fault	Stop
R-15	V60 at or above 1.50 mm/s with every limit clear	L1 watch [I]	P60, PMR300, HB300, I60	Early F02 or early F01	Trend; check pressure first
R-16	P60 at or below 4.30 bar with every limit clear	L1 watch [I]	V60 and HB300 rising?	Early F01	Trend; plan filter inspection

Table 7. Threshold classification table

Channel	L0 Normal	L1 Watch [I]	L2 Warning [P]	L3 Alarm [P]	L4 Critical [I]	Unit
Ipk	up to 153	above 160	at or above 175	at or above 195	at or above 220	A
I60	90 to 99	above 99.5	not defined	not defined	not defined	A
P60 / P10	4.35 to 4.65	at or below 4.30	at or below 3.6	at or below 3.0	at or below 2.5	bar
V60 / V10	1.2 to 1.45 (V60)	at or above 1.50	at or above 2.5	at or above 4.0	at or above 7.0	mm/s
HB300	0.03	at or above 0.12	at or above 0.24	not defined	not defined	mm/s

4. Steps 4 to 7: Persistence, Rate, State and Load

4.1 Step 4: persistence and duration

Table 8. Persistence rules

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-28	Value beyond a limit for 10 s (open) or inside for 10 s (clear)	Persistence	Window means	Level change	Apply to every level [P] planned
R-35	Single sample outside a limit with window mean inside	Transient	Next 10 s	Spike or data error	Do not alarm

Table 9. Why persistence is needed for each channel

Channel	Cyclic content	Effect on per-sample status	Window used
V	Breathes with the 4.6 s stroke	BEARING_WEAR at r 0.6 to 0.8 flickers across 2.5 mm/s (2 percent to 88 percent of samples)	10 s mean for status, PF60 for flicker [D]
P	1 Hz pulsation, 0.03 bar noise	Near 3.6 bar the status flickers (P at r = 0.44 has mean 3.61 and range 3.48 to 3.72)	10 s mean
I	Working-stroke pulse and 4 A ripple	Single samples hit 150 A in every stroke	lpk over 3 strokes

4.2 Step 5: rate of change

Table 10. Rate rules

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-21	V60 climbs steadily from about 1.3 toward 3 mm/s over hours	Drift	P60, shape	F02 or F01	Treat as drift whatever the limit says [P]
R-27	P60 falls by 0.3 bar or V60 rises by 0.3 mm/s within 10 minutes [I]	Drift L1	Both features	F01 or F02	Open a watch investigation
R-25	Delta V60 / Delta P60 about -1.8 mm/s per bar over 10 minutes	Consistent with clog	HB300	F01 consistency pass	Raise confidence
R-29	Estimated time to alarm below 15 minutes of real time [I]	Escalation +1	60 s slope	Rapid progression	Raise one level

Table 11. Rate thresholds at a glance [D] / [I]

Feature	Healthy slope	Drift L1	Rapid (escalate)
P60	0 bar/min	Falls 0.3 bar in 10 minutes	Estimated time to 3.0 bar under 15 minutes
V60	0 mm/s/min	Rises 0.3 mm/s in 10 minutes	Estimated time to 4.0 mm/s under 15 minutes
I60	0 A/min	Rises 2 A in 10 minutes	n/a

Time to alarm = (limit - current value) / slope of the 60 s mean over the last 10 minutes. For the model at rampSec 120 s the clog slopes are -1.01 bar/min and +1.84 mm/s/min [D]; at 600 s they are one fifth of that. The repository planned detector also uses a rolling z-score above 3 for 30 s or a sustained positive slope of the 60 s mean over 10 minutes [P] planned.

4.3 Steps 6 and 7: operating state and load

Table 12. State and load rules

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-26	Current evaluation	Stroke gating	Use Ipk over at least 3 strokes (13.8 s)	Single samples are not diagnostic	Never judge on one sample
R-23	Ipk at or above 160 A with idle current about 88 A and V60, P60 normal	Process load	Die, material	F09	Check die setup
R-31	Machine stopped, in setup or die change [E]	Suppress	State input	No limit-based diagnosis	Wait for production

Table 13. Load attribution

Observation	Attribution	Reason
Ipk up, idle 88 A unchanged, V60 and P60 normal	Process load (F09)	Increase is confined to the working stroke
I60 and idle both up, V60 up	Friction or bearing (F02, F01) or motor (F08)	Whole-cycle change
V up only in the 62.5 to 150 Hz structural band, in the working stroke	Impact or process	Band swells with load [P]
P changes	Never load	Pressure has no stroke-phase term [P]

5. Steps 8 and 9: Candidate Generation and Differentiation

5.1 Step 8: candidate generation

Table 14. Candidate generation from the leading observation

Leading observation	Candidates (in order)	Reason
V10 at or above 2.5 mm/s, P60 falling	F01, F16, F03, F04	Lubrication faults lower the pressure
V10 at or above 2.5 mm/s, P60 normal	F02, F10, F11, F13	No pressure change
P10 at or below 3.6 bar, V60 normal	F12, F03, F04, F06, F01 early	Vibration has not followed
P rising	F05, F06, F07	Different fault [P]
l _{pk} at or above 175 A	F09, F08, F14	Current limit
V60 at 1.5 to 2.4 mm/s with no limit	F02 early, F01 early, F10	Watch level
Flat or implausible channel	F12, F13, F14, F15	Data first

5.2 Evidence-weight matrix

Each piece of evidence adds points to every candidate (negative points contradict it). The scores are prototype values [I]; the evidence definitions come from the repository signatures [P].

Table 15. Evidence-weight matrix

ID	Evidence	F01	F02	F03/04	F09	F12	F13	F16
E1	P60 at or below 4.30 bar and falling	+3	-3	+3	-2	+1	-1	+3
E2	P60 within 4.45 to 4.55 bar	-3	+2	-3	+1	0	+1	-3
E3	V60 at or above 1.50 mm/s and rising	+2	+2	+1	0	-2	+1	+2
E4	HB300 at or above 0.24 mm/s with PMR300 below 2.5 (flat floor)	+3	-2	+1	-1	-3	0	0
E5	Discrete non-integer lines: PMR300 above 3, defect bins at least 4 x healthy	-3	+4	-2	-1	-1	-1	+3
E5n	No discrete lines (PMR300 below 2.5)	+1	-3	+1	+1	0	0	-3
E6	V60 at or above 4.0 mm/s	+2	-3	+1	0	-1	+1	+2
E7	Delta I60 / Delta V60 about 0.8 A per mm/s	+2	-2	0	-1	0	0	+1
E8	Delta I60 / Delta V60 about 4.8 A per mm/s	-2	+3	-1	+1	0	-1	+1
E9	l _{pk} at or above 160 A with idle level about 88 A	-1	-1	-1	+4	0	0	-1
E10	RMS differs from the spectrum root-sum-square, or V flat or implausible	-3	-3	-2	-2	0	+4	-3
E11	Pressure ripple 0.12 bar missing or P flat	-2	0	+1	0	+4	0	-2
E12	Step change instead of a ramp	-2	-2	-1	+1	+2	+3	-2
E13	P falling and discrete lines both present	-1	-1	0	-2	-1	-1	+4
E14	V60 and HB300 do not move with P (P fall alone)	-3	0	+2	0	+3	0	-3

Table 16. Decision from scores [I]

Quantity	Rule	Meaning
Margin 4 or more	Top score minus second	Clear: act on the top candidate
Margin 2 to 3	Top minus second	Ambiguous: apply R-20 (lubrication first) and gather more evidence
Margin 0 to 1	Top minus second	Unresolved: inspection only; no corrective action
Top score 9 or more	Confidence	High
Top score 6 to 8	Confidence	Medium
Top score 3 to 5	Confidence	Low

5.3 Differentiation rules

Table 17. Fault differentiation rules

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-17	HB300 at or above 0.24 mm/s and PMR300 below 2.5	Flat floor evidence	P60	F01 (broadband floor above 300 Hz)	Lubrication first
R-18	PMR300 above 3 and defect bins 7, 14, 21, 28, 35 at least 4 x healthy	Discrete family evidence	P60 normal?	F02 bearing wear	Handheld analyser; plan replacement
R-19	Pressure normal and the spectrum shows discrete peaks	Not a filter problem	Bearing procedure	F02	Use the bearing runbook [P]
R-20	Evidence is mixed between a lubrication fault and a bearing fault	Lubrication first	Filter and pressure trend	F01 or F16	Replace the filter first, then re-check the spectrum [P]
R-22	P60 rises above 4.65 bar, or above 5.0 bar [I]	Different fault	V60, state, temperature	F05 or F06 (not modelled)	Inspect after the transducer; not a filter problem [P]
R-24	Delta I60 / Delta V60 above 3.5 A per mm/s	Bearing or electrical	Lines, line-frequency content	F02 or F08	Bearing inspection; motor test if line lines
R-34	Transducer-like signature: P flat or offset with V and HB300 unchanged	Sensor suspect	Reference gauge	F12	Verify sensor before filter work

6. Decision Matrix and Inspection Priority

Table 18. Diagnostic decision matrix: observation to action

Observation	Classification	Check	Diagnosis	Action
P60 falling, V60 rising, flat floor, dl/dV about 0.8	L2 to L3	Filter cartridge, line pressure	F01	Replace cartridge
V60 rising, P60 normal, lines at 88, 175, 263, 351, 439 Hz	L1 to L2	Handheld analyser	F02	Plan bearing replacement
V60 at 3 mm/s, P60 3.9 bar, floor with some lines	L2, ambiguous	Filter first	F16 or F01	Replace filter; re-check spectrum
P low, V normal, ripple missing	LS	Transducer vs reference	F12	Fix sensor
l _{pk} 180 A, idle 88 A	L2	Die and process	F09	Check setup
V10 8 mm/s, P and I normal, RMS differs from spectrum	LS / L4	Sensor mount	F13	Refit sensor
P rising to 5.5 bar	Other	Lines after transducer	F05 or F06	Inspect
stale, offline	LS	diag	F15	Fix link

6.1 Inspection priority

Table 19. Inspection prioritisation

Priority	Item	Why first	Applies when
1	Data path and sensor against a reference	Cheapest and removes false alarms	Any LS or flat signal
2	Lube filter and line pressure	Cheap fault that causes bearing damage if left [P]	P60 falling or mixed evidence
3	Lube oil level, pump, leaks	Next lubrication causes	Pressure falls but filter swap does not help
4	Bearing frequencies with a handheld analyser	Confirms outer-race family	Pressure normal and lines present
5	Motor, supply and drive	Electrical causes	Current ratio above 4.8 A per mm/s or line lines
6	Die, alignment, fasteners	Process and mechanical looseness	l _{pk} up with idle normal; integer orders

7. Steps 11 and 12: Severity and Escalation

Table 20. Severity classification and escalation matrix

Level	Severity	Notify	Time to act [I]	Condition for stop
L1 Watch	S1 to S2	Planner	Next planned inspection	None
L2 Warning	S2	Technician	Next planned stop	None
L3 Alarm	S3	Supervisor and maintenance	This shift	Supervisor decides
L4 Critical	S4	Manager	Immediate	Controlled stop
LS	S1	Instrumentation / IT	Within shift	Not a machine stop

Table 21. Escalation modifiers

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-29	Estimated time to alarm below 15 minutes of real time [I]	Escalation +1	60 s slope	Rapid progression	Raise one level
R-30	Two channels at L2 or above	Escalation +1	Both channels	Compound condition	Raise one level (max L4)

Table 22. Severity score [I]

Component	Points	Condition
Level	1 to 4	L1 to L4
Progression	+1	Estimated time to alarm below 15 minutes
Compound	+1	Two channels at L2 or above
Fault class	+1	F02, F03, F04, F08 or F16 (high consequence)
Data suspect	cap 1	LS

Severity score 1 to 2 is S1 or S2, 3 to 4 is S3, 5 or more is S4. The repository's consequence statements anchor the classes: a clogged filter is a short cartridge swap, a bearing that fails in service costs days [P].

8. Steps 13 and 14: Corrective Action and Verification

Table 23. Corrective action by diagnosis

Diagnosis	Action	Safety condition	Source
F01	Replace the filter cartridge (a short task, in the order of minutes)	OEM procedure; lock-out and tag-out; relieve line pressure; specified cartridge	[P]
F02	Inspect at next planned stop with a handheld analyser; plan bearing replacement	OEM procedure; lock-out and tag-out before opening the main drive; specified bearings and fits	[P]
F03, F04	Repair pump or leak; refill with the specified oil	OEM procedure	[E]
F05, F06	Clear restriction or service the valve	OEM procedure	[E]
F08, F09	Electrical test or process correction	Qualified electrician	[E]
F10, F11	Align, balance or tighten	OEM procedure	[E]
F12 to F15	Sensor or data repair	Instrument technician	[E]

8.1 Post-maintenance verification criteria

Table 24. Verification criteria after a repair [P] / [I]

Quantity	Pass	Fail means	Source
P60	4.35 to 4.65 bar, ripple present	Pressure not restored: other lubrication cause or transducer	[P] acceptance about 4.5 bar
V60	1.20 to 1.45 mm/s (baseline about 1.3)	Residual defect or another fault	[P] / [I]
HB300	At most 0.06 mm/s (twice healthy)	Floor remains	[I]
PMR300 and defect bins	Defect bins below 4 x healthy	Bearing defect remains	[P] test
I60	Within 3 percent of 94.8 A	Load or friction remains	[I]
Duration	60 minutes of steady running with no L1 or above [I]	Reopen investigation	[I]

Table 25. Verification rule

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-32	After a repair	Verify	P60 4.35 to 4.65 bar, V60 1.2 to 1.45 mm/s, HB300 at most 0.06 mm/s, I60 within 3 percent of baseline	Repair successful or not	Return to service or reopen
R-33	Report is produced	Content rule	Observation, diagnosis, evidence, action, confidence, citations, OEM and lock-out caveat	Planned guard [P]	Reject reports without the caveat

8.2 Confidence and specialist conditions

Table 26. Conditions that require a specialist

Condition	Reason	Who
Margin below 4 after lubrication check	Unresolved diagnosis	Vibration analyst
Bearing lines confirmed	Replacement planning	Maintenance engineer
L4 or rapid progression	Safety	Production manager
Electrical suspicion	Qualified work	Electrician
LS persistent	Instrument or network	Instrumentation / IT

9. Worked Examples

Examples A and B use project numbers [P] / [D] from the derived tables. Examples C and D are illustrative [I] because the repository models neither a combined fault nor a transducer fault. Scores come from the evidence-weight matrix.

9.1 Example A: clogged filter at mid severity

Table 27. Example A: clogged filter at mid severity: observation

Item	Value
State	Running; status online; no gaps; data valid (R-01 to R-04 pass)
P60 / P10	3.49 bar; P10 reached 3.6 bar earlier and stays below it [D] ($r = 0.5$)
V60	3.15 mm/s (warn reached, 100 percent of samples at or over 2.5) [D]
HB300 / PMR300	0.447 mm/s (15 x healthy) / below 2.5 [D] / [P]
I60	96.2 A: ΔI over $\Delta V = 1.4 \text{ A} / 1.84 \text{ mm/s} = 0.76 \text{ A per mm/s}$ [D]
Level	P: L2; V: L2; two channels so escalation +1 to L3
Diagnosis	F01 CLOGGED_FILTER. Action: replace the cartridge; verify P60 4.35 to 4.65 bar and V60 1.2 to 1.45 mm/s

Table 28. Example A: clogged filter at mid severity: evidence and scores

Evidence matched	Scores (F01 / F02 / F03-04 / F09 / F12 / F13 / F16)	Result
E1, E3, E4, E5n, E7	11 / -8 / 6 / -3 / -4 / 0 / 3	Top F01 (11), second F03/04 (6), margin 5 (clear), confidence High

9.2 Example B: early bearing wear below every limit

Table 29. Example B: early bearing wear below every limit: observation

Item	Value
State	Running; data valid
P60	4.50 bar (unchanged)
V60	1.94 mm/s (r about 0.4: $1.31 + 1.58 \times 0.4$) [D]; no sample at or over 2.5 mm/s; platform status normal
I60	97.8 A (+3.2 percent) [D]: ΔI over $\Delta V = 3.0 \text{ A} / 0.63 \text{ mm/s} = 4.8 \text{ A per mm/s}$
Spectrum	Discrete lines at 88, 175, 263, 351, 439 Hz at least 4 x healthy; PMR300 above 3 [P]
Level	V60 L1 (watch); no limit; drift
Diagnosis	F02 BEARING_WEAR early. Action: inspection at the next planned stop with a handheld analyser; plan replacement. A threshold alarm alone never fires [P]

Table 30. Example B: early bearing wear below every limit: evidence and scores

Evidence matched	Scores (F01 / F02 / F03-04 / F09 / F12 / F13 / F16)	Result
E2, E3, E5, E8	-6 / 11 / -5 / 1 / -3 / 0 / 3	Top F02 (11), second F16 (3), margin 8 (clear), confidence High

9.3 Example C: mixed evidence (illustrative)

Table 31. Example C: mixed evidence (illustrative): observation

Item	Value
State	Running; data valid
P60	3.9 bar and falling [I]
V60	3.2 mm/s [I]
Spectrum	Raised floor above 300 Hz (HB300 0.35 mm/s) and some discrete lines (PMR300 2.8) [I]
Level	P: L1; V: L2
Diagnosis	Ambiguous between F16 and F01: apply R-20. Replace the filter first, then re-check the spectrum; if lines persist, plan bearing inspection

Table 32. Example C: mixed evidence (illustrative): evidence and scores

Evidence matched	Scores (F01 / F02 / F03-04 / F09 / F12 / F13 / F16)	Result
E1, E3, E5, E4	5 / 1 / 3 / -4 / -5 / -1 / 8	Top F16 (8), second F01 (5), margin 3 (ambiguous), confidence Medium

9.4 Example D: flat pressure with healthy vibration (illustrative)

Table 33. Example D: flat pressure with healthy vibration (illustrative): observation

Item	Value
State	Running; data valid
P60	3.20 bar with standard deviation near zero for 5 minutes [I] (healthy ripple 0.12 bar missing)
V60 / HB300	1.30 mm/s / 0.03 mm/s [I]
Level	P: L2 on raw limit; but no corroboration
Diagnosis	F12 pressure transducer fault suspected. Action: compare with a reference gauge before touching the filter

Table 34. Example D: flat pressure with healthy vibration (illustrative): evidence and scores

Evidence matched	Scores (F01 / F02 / F03-04 / F09 / F12 / F13 / F16)	Result
E11, E14, E2	-8 / 2 / 0 / 1 / 7 / 1 / -8	Top F12 (7), second F02 (2), margin 5 (clear), confidence Medium

10. Checklists, Report Template and Pseudo-code

10.1 Checklists

Table 35. Pre-diagnosis checklist

Check	Done when
Data valid	status online, no gaps, RMS matches spectrum
State known	Machine running, not in setup
Baseline available	Healthy or recorded baseline
Window complete	At least 60 s of data
No injected event	No ack of cmd/scenario in the window

Table 36. Pre-maintenance checklist

Check	Done when
Diagnosis confidence	Medium or High, or lubrication-first rule applied
Safety	Lock-out and tag-out plan, OEM procedure available
Parts	Specified cartridge or bearings identified from the OEM documents
Verification plan	Criteria of the verification table

10.2 Report template

Table 37. Report fields [P] planned

Field	Content
Observation	Channel, value, window, level, persistence
Diagnosis	Fault ID and name, confidence tier, score margin
Evidence	Features and their values against the baseline
Recommended action	Action, urgency, verification criteria
Confidence	Computed in code from signature matches, never from the model's impression
Citations	Document and section identifiers
Caveat	Follow the OEM procedure and apply lock-out and tag-out; the agent cannot act on the machine

10.3 Pseudo-code

Table 38. Decision pseudo-code [I]

<pre> if not valid(data): return LS(reason) # R-01 to R-04 if state in (stopped, setup): return SUPPRESSED # R-31 levels = {ch: level(window_mean(ch)) for ch in channels} # R-06 to R-16 if all(l == L0 for l in levels.values()) and no drift: return L0 cands = generate(levels, features) # Section 7 scores = sum_evidence(cands) # evidence-weight matrix top, second = rank(scores) if top - second < 2: apply R-20 (lubrication first) severity = score(levels, rate, compound, fault_class) return report(observation, top, evidence, action, confidence, citations) </pre>
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11. Edge Cases and Special Situations

Table 39. Edge cases

Situation	Treatment	Reason
V flickers across 2.5 mm/s with V60 below 2.5	Use PF60; treat as L1 if PF60 above 0, L2 if V60 reaches 2.5	BEARING_WEAR flicker at r 0.6 to 0.8 [P]
P crosses 3.6 bar only on single samples	No alarm; check P10	Noise 0.03 bar and ripple 0.12 bar
Both channels warn simultaneously	Compound, lubrication first	Clog warns on V before P
Fault ramp very fast (5 s)	Rules operate on 10 s windows: expect L3 within 10 s of the ramp end	Ramp allowed 5 to 600 s [P]
Fault ramp very slow (600 s)	Drift rules and slope matter; limits come late	Slope one fifth of default
Reboot clears the fault	Treat as reset; recompute baseline	cmd/reboot [P]
Spectrum older than 2 s	Use latest; do not mix with older RMS	spectrum interval
Current high only at one stroke	Do not alarm	R-26
P rising	Not a filter fault	R-22

12. Level Playbooks

Table 40. Playbooks

Level	Immediate	Within the shift	Closure
L1	Open a watch investigation	Trend; confirm features; plan inspection	Close if features return inside the watch band for 60 minutes
L2	Notify technician; run Steps 8 and 9	Plan work at next stop; lubrication first if mixed	Verify after repair
L3	Notify supervisor	Perform filter swap or inspection; decide on stop	Verify after repair
L4	Controlled stop	Inspect; consult OEM	Return-to-service review
LS	Notify instrumentation / IT	Resolve data path or sensor	Data valid for 60 minutes

13. Consolidated Rule Library

All IF/THEN rules used in this document. Rule identifiers are cited in the report so that a reader can find the rule here.

Table 41. Rules R-01 to R-18

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-01	status is not online, or no telemetry for max(5 s, 4 x intervalMs)	LS data state	gateway diag (uptimeS, rssi, mqttReconnects), status topic	F15 data-path fault	No mechanical diagnosis; follow the data-path table
R-02	bootId changes with seq back at 0	LS information	uptimeS, reset reason, recent cmd/reboot	Gateway reboot (a simulated fault is cleared by cmd/reboot)	Restart the window and recompute the baseline
R-03	Standard deviation of a channel is near zero over 60 s while the others vary	LS	Healthy P has 0.03 bar noise and 0.12 bar ripple; I and V breathe with the stroke	F12 (P), F13 (V) or F14 (I)	Inspect the sensor against a reference
R-04	Received RMS differs from the square root of the summed squared spectrum bins by more than 2 percent [I]	LS	Latest spectrum within 2 s; decoding	F13 or F15	Resolve data before any mechanical diagnosis
R-05	An ack with ok true for cmd/scenario was seen in the window	Injected simulation event	ack detail, cmdId	Operator fault injection	Record as injected; do not escalate as a plant fault
R-06	P10 at or below 3.6 bar for 10 s	L2 warning	V60, HB300, PMR300, I60 ratio	F01 candidate (also F03, F04, F06, F12)	Run differentiation; replace filter if F01 confirmed
R-07	P10 at or below 3.0 bar for 10 s	L3 alarm	V60 at or above 4.0 mm/s and a broadband floor	F01 severe, or F03, F04	Intervene this shift; filter swap if broadband
R-08	P10 at or below 2.5 bar for 10 s [I]	L4 critical	V10, I60, state	Oil starvation of the bearings	Controlled stop and inspect
R-09	V10 at or above 2.5 mm/s for 10 s	L2 warning	P60, PMR300, HB300	F01 if P60 falls, F02 if P60 normal with lines	Lubrication first; inspect at next stop
R-10	V10 at or above 4.0 mm/s for 10 s	L3 alarm	P60 at or below 3.0 bar?	F01 (bearing wear never reaches 4.0 mm/s in the model [P])	Filter swap now; inspect bearings afterwards
R-11	V10 at or above 7.0 mm/s for 10 s [I]	L4 critical	Mount and cable, impacts, other channels	Severe fault or F13	Controlled stop; check sensor
R-12	Ipk at or above 175 A in 2 of 3 strokes	L2 warning	Idle level 88 A? V60 and P60 normal?	F09 process or F08 motor	Check die, process, drive
R-13	Ipk at or above 195 A in 2 of 3 strokes	L3 alarm	Stroke profile, drive protection	F09 or F08 severe	Reduce load; inspect
R-14	Ipk at or above 220 A, or a sample at or above 240 A [I]	L4 critical	Jam or short	Overload or electrical fault	Stop
R-15	V60 at or above 1.50 mm/s with every limit clear	L1 watch [I]	P60, PMR300, HB300, I60	Early F02 or early F01	Trend; check pressure first
R-16	P60 at or below 4.30 bar with every limit clear	L1 watch [I]	V60 and HB300 rising?	Early F01	Trend; plan filter inspection
R-17	HB300 at or above 0.24 mm/s and PMR300 below 2.5	Flat floor evidence	P60	F01 (broadband floor above 300 Hz)	Lubrication first
R-18	PMR300 above 3 and defect bins 7, 14, 21, 28, 35 at least 4 x healthy	Discrete family evidence	P60 normal?	F02 bearing wear	Handheld analyser; plan replacement

Table 42. Rules R-19 to R-35

Rule	IF condition	THEN classification	CHECK	Possible diagnosis	Recommended action
R-19	Pressure normal and the spectrum shows discrete peaks	Not a filter problem	Bearing procedure	F02	Use the bearing runbook [P]
R-20	Evidence is mixed between a lubrication fault and a bearing fault	Lubrication first	Filter and pressure trend	F01 or F16	Replace the filter first, then re-check the spectrum [P]
R-21	V60 climbs steadily from about 1.3 toward 3 mm/s over hours	Drift	P60, shape	F02 or F01	Treat as drift whatever the limit says [P]
R-22	P60 rises above 4.65 bar, or above 5.0 bar [I]	Different fault	V60, state, temperature	F05 or F06 (not modelled)	Inspect after the transducer; not a filter problem [P]
R-23	Ip _k at or above 160 A with idle current about 88 A and V60, P60 normal	Process load	Die, material	F09	Check die setup
R-24	Delta I60 / Delta V60 above 3.5 A per mm/s	Bearing or electrical	Lines, line-frequency content	F02 or F08	Bearing inspection; motor test if line lines
R-25	Delta V60 / Delta P60 about -1.8 mm/s per bar over 10 minutes	Consistent with clog	HB300	F01 consistency pass	Raise confidence
R-26	Current evaluation	Stroke gating	Use Ip _k over at least 3 strokes (13.8 s)	Single samples are not diagnostic	Never judge on one sample
R-27	P60 falls by 0.3 bar or V60 rises by 0.3 mm/s within 10 minutes [I]	Drift L1	Both features	F01 or F02	Open a watch investigation
R-28	Value beyond a limit for 10 s (open) or inside for 10 s (clear)	Persistence	Window means	Level change	Apply to every level [P] planned
R-29	Estimated time to alarm below 15 minutes of real time [I]	Escalation +1	60 s slope	Rapid progression	Raise one level
R-30	Two channels at L2 or above	Escalation +1	Both channels	Compound condition	Raise one level (max L4)
R-31	Machine stopped, in setup or die change [E]	Suppress	State input	No limit-based diagnosis	Wait for production
R-32	After a repair	Verify	P60 4.35 to 4.65 bar, V60 1.2 to 1.45 mm/s, HB300 at most 0.06 mm/s, I60 within 3 percent of baseline	Repair successful or not	Return to service or reopen
R-33	Report is produced	Content rule	Observation, diagnosis, evidence, action, confidence, citations, OEM and lock-out caveat	Planned guard [P]	Reject reports without the caveat
R-34	Transducer-like signature: P flat or offset with V and HB300 unchanged	Sensor suspect	Reference gauge	F12	Verify sensor before filter work
R-35	Single sample outside a limit with window mean inside	Transient	Next 10 s	Spike or data error	Do not alarm

14. Reasoning Chains

Table 43. Reasoning chains: sensor reading to verification

Chain	Sensor reading	Abnormal condition	Affected component	Possible fault	Differentiation	Inspection	Action	Verification
Lubrication	P10 at 3.4 bar, V10 2.9 mm/s	Warning on P and V	Lube filter, main bearing housing	F01	Flat floor, dl/dV 0.8	Cartridge	Replace	P60 4.5 bar, V60 1.3
Bearing	V60 2.1 mm/s, P normal	Watch to warning	Main bearing housing	F02	Lines at defect orders, dl/dV 4.8	Handheld analyser	Plan replacement	Lines gone, V60 1.3
Process	IpK 182 A	Warning on current	Press, drive	F09	Idle 88 A	Die	Correct	IpK about 150 A
Sensor	P flat 3.2 bar	LS	LUBE_OIL_PRE SSURE sensor	F12	V, HB300 unchanged	Reference gauge	Repair sensor	P ripple back
Data	stale	LS	Gateway	F15	diag fields	Network	Fix link	status online

15. Summary, Safety, References and Revision

15.1 Summary

- Validate data first, classify levels on 10 s and 60 s windows, then generate candidates and score them with the evidence-weight matrix.
- Pressure, spectrum shape, current ratio and alarm behaviour separate the two simulated faults; mixed evidence means lubrication first [P].
- Never rely on a threshold alarm alone: BEARING_WEAR never reaches 4.0 mm/s [P].

15.2 Key thresholds

Table 44. Key thresholds

Quantity	Value	Tag
P warn / alarm / critical	3.6 / 3.0 bar / 2.5 bar	[P] / [P] / [I]
V warn / alarm / critical	2.5 / 4.0 / 7.0 mm/s	[P] / [P] / [I]
I warn / alarm / critical	175 / 195 / 220 A	[P] / [P] / [I]
Persistence	10 s open, 10 s clear	[P] planned
Margin rule	4 clear, 2 to 3 ambiguous, 0 to 1 unresolved	[I]

15.3 Safety and escalation notes

This document is a capstone prototype and not an OEM manual. Every maintenance recommendation requires the manufacturer's procedure and lock-out and tag-out. The agent cannot act on the machine; it reports.

15.4 References and engineering basis

- CONTEXT_PRESS_STAMP_01.md; docs/runbooks/press-lubrication-circuit.md; docs/runbooks/press-main-bearings-vibration.md; docs/phase5-agent-blueprint.md (planned detector and signatures) [P].
- Evidence-weight scores, margins and confidence tiers are illustrative and must be calibrated with recorded cases [I].

15.5 Revision notes

Table 45. Revision history

Rev	Date	Change
1.0	2026-10-08	First issue